

Project Plan and Evaluation Summary^[4pt] Context-Aware False Alarm Reduction for Violence Detection in Surveillance Videos Using a Lightweight Gating Module^[8pt] Group 2 DAPM391 – SU26 FPT University

1. Introduction and Problem Statement

The objective of this project is to build a practical CCTV violence detection system that reduces the False Positive Rate (FPR) without increasing the False Negative Rate (FNR). Current deep learning models such as X3D can recognize violent actions, but they often produce false alarms in complex real-world surveillance scenes, including crowded areas, bars, sports events, low-light environments, and blurred footage.

To address this issue, the project integrates an X3D-S backbone with a lightweight Context Gating Module (CGM). Instead of only deciding whether a clip looks violent, the system also evaluates whether the surrounding context makes the base prediction reliable.

Core Idea

The proposed system combines action recognition with situational awareness. X3D-S provides the base violence probability, while the Context Gating Module uses crowd, lighting, and motion-chaos cues to adjust unreliable predictions and reduce false alarms.

2. Proposed Pipeline

The proposed pipeline contains three main tiers.

Tier 1: Base Violence Detector

The first tier uses X3D-S as the base violence detector. It analyzes RGB video clips and outputs a base violence probability, denoted as p_{base} . After the base detector is trained, the backbone is frozen so that the next stage can focus on learning the effect of context features.

Tier 2: Context Streams

The second tier extracts annotation-free context features from each video clip. The project uses three context streams:

- **Crowd stream:** uses YOLOv8n to estimate crowd density and human presence.
- **Lighting stream:** uses OpenCV to compute brightness, contrast, blur level, and low-light regions.
- **Motion-chaos stream:** uses Farneback Optical Flow to measure scene-level motion intensity, entropy, and synchrony.

Together, these streams form 12 context features without requiring manual labeling.

Tier 3: Context Gating Module

The final tier is the Context Gating Module. A 13-dimensional input vector is created by combining p_{base} with the 12 context features. This vector is passed into lightweight MLPs to estimate a confidence weight α and a context-calibrated probability p_{ctx} .

The final prediction is computed as:

$$p_{final} = \alpha \cdot p_{base} + (1 - \alpha) \cdot p_{ctx}$$

Here, α controls how much the system should trust the base detector. A low α means the context module has stronger influence on the final decision.

3. Research Questions and Evaluation Summary

RQ1: Does X3D-S + CGM reduce FPR compared with X3D-S alone?

This question evaluates whether the proposed CGM can reduce false positives while keeping FNR stable. The main focus is practical reliability in CCTV deployment, not only overall accuracy. The CGM is designed to correct sensitive false alarms where $p_{base} \geq 0.5$ by using context features that are not strongly correlated with the base prediction.

RQ2: Which context stream contributes the most to false-alarm reduction?

This question studies the contribution of each context stream. The full CGM using all three streams is expected to perform best because crowd, lighting, and motion-chaos features complement each other. Individually, the Lighting stream is expected to be the strongest contributor for separating violent and non-violent clips, based on metrics such as Cohen’s d and AUC.

RQ3: Does the model generalize across datasets?

This question evaluates cross-dataset generalization from RWF-2000 to RLVS. Since the two datasets have different video lengths, resolutions, and scene conditions, strong zero-shot performance would indicate that the context gate can reduce FPR even under domain shift.

RQ4: Is the Motion-chaos stream redundant alongside X3D?

This question checks whether the motion stream provides information that X3D does not already capture. X3D learns motion at the action level, while the Motion-chaos stream measures scene-level entropy and synchrony. Therefore, it helps distinguish real fights from safe but intense movements such as sports, synchronized running, crowd movement, or camera shake.

4. Weekly Project Plan

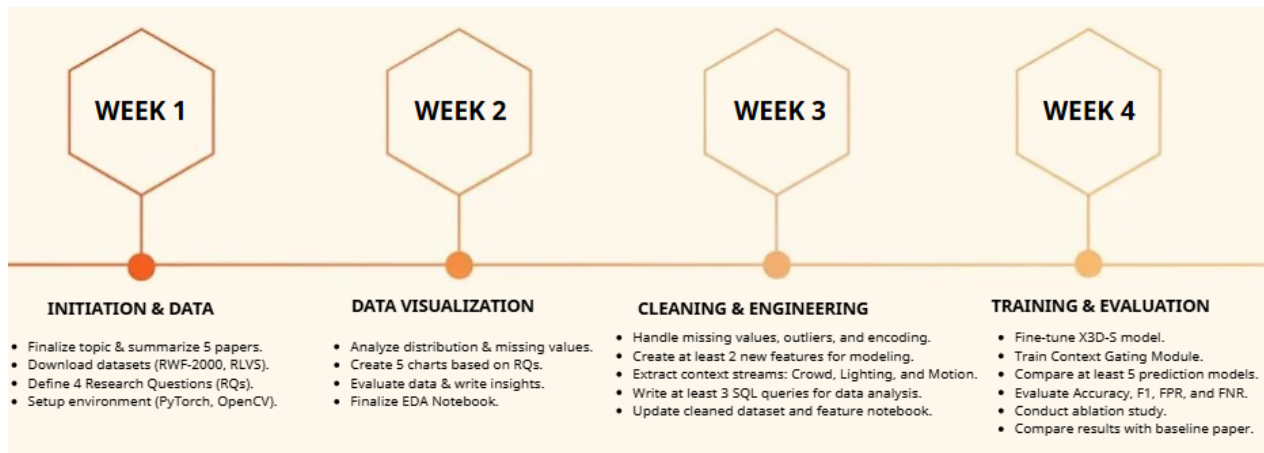


Figure 1: Four-week project timeline.

Week 1: Project Initialization and Data Preparation

The goal of Week 1 is to finalize the project topic, define the datasets, create the project plan, and establish the research questions. The group also reviews at least five related papers, downloads RWF-2000 and RLVS, checks basic dataset integrity, and sets up the working environment with Python 3.10, PyTorch 2.x, CUDA, ultralytics, and OpenCV.

Main outputs: finalized topic, research questions, related work summary, verified datasets, and working environment.

Week 2: Data Visualization and Exploratory Data Analysis

The goal of Week 2 is to perform EDA and build visualizations that directly support the research questions. The group analyzes dataset structure, class distribution, missing values, and data quality issues. At least five visualizations are created, and each chart must include analytical insights rather than only visual description.

Main outputs: EDA notebook, dataset statistics, visualization results, and written insights.

Week 3: Data Cleaning and Feature Engineering

The goal of Week 3 is to clean the dataset and extract useful features for later modeling. The group handles missing values, outliers, and encoding issues where necessary. At least two new features are created. The three context streams are also extracted: Crowd using YOLOv8n, Lighting using OpenCV, and Motion-chaos using Farneback Optical Flow. At least three meaningful SQL queries are written for data analysis.

Main outputs: cleaned dataset, engineered features, context streams, SQL analysis, and updated notebook.

Week 4: Model Training and Evaluation

The goal of Week 4 is to train and evaluate the models. The group fine-tunes X3D-S, trains the Context Gating Module, compares at least five prediction models, and evaluates them

using Accuracy, F1-score, FPR, and FNR. Ablation studies from E0 to E4 are conducted to measure the contribution of each component. The final report and presentation slides are also completed.

Main outputs: trained models, metric comparison, ablation study, final report, and slides.

5. Member Responsibilities

Hà Việt Hưng

Hà Việt Hưng is responsible for creating the project plan, defining the research questions, managing the technical phases, performing feature engineering, extracting the Motion-chaos stream, fine-tuning X3D-S, and setting up the Context Gating Module.

Nguyễn Viết Nhân

Nguyễn Viết Nhân is responsible for downloading and verifying RWF-2000 and RLVS, creating basic data visualizations, extracting Crowd features using YOLOv8n, extracting Lighting features using OpenCV, and training additional prediction models for comparison.

Nguyễn Thái Kiệt

Nguyễn Thái Kiệt is responsible for reading and summarizing related papers, designing advanced visualizations, handling missing values, outliers, and encoding with Scikit-learn, and preparing the final comparison of evaluation metrics.

Trần Bảo Nguyên

Trần Bảo Nguyên is responsible for setting up the shared environment, extracting analytical insights from visualizations, writing SQL queries for data analysis, running ablation studies, analyzing model performance differences, and preparing the final presentation slides.

6. Expected Deliverables

By the end of the project, the group is expected to deliver:

- A finalized project topic, project plan, and four research questions.
- A related work summary covering at least five scientific papers.
- Verified RWF-2000 and RLVS datasets.
- A Python notebook for EDA, visualization, cleaning, and feature engineering.
- Extracted Crowd, Lighting, and Motion-chaos context features.
- A fine-tuned X3D-S baseline model.
- A lightweight Context Gating Module for false-alarm reduction.
- A comparison of at least five models using Accuracy, F1-score, FPR, and FNR.
- Ablation study results showing the contribution of each context stream.
- A final report and presentation slides.

7. Conclusion and Deployment Value

The key contribution of this project is an end-to-end violence detection pipeline that reduces false alarms without requiring manual labels for context features. The system is also interpretable for real-world deployment because suppressed alerts can be explained through logs such as low α , high blur, low lighting, or dense crowd conditions.

Overall, the project shows that situational awareness is an important bridge between academic action recognition models and practical surveillance systems.